arXiv:2509.09396v1 [cs.LG] 11 Sep 2025

LLMs Don't Know Their Own Decision Boundaries: The Unreliability of Self-Generated Counterfactual Explanations

Harry Mayne[1], Ryan Othniel Kearns[1], Yushi Yang[1], Andrew M. Bean[1], Eoin Delaney[2], Chris Russell[1], Adam Mahdi[1]

[1]University of Oxford, [2]Trinity College Dublin

Abstract

To collaborate effectively with humans, language models must be able to explain their decisions in natural language. We study a specific type of self-explanation: *self-generated counterfactual explanations* (SCEs), where a model explains its prediction by modifying the input such that it would have predicted a different outcome. We evaluate whether LLMs can produce SCEs that are *valid*, achieving the intended outcome, and *minimal*, modifying the input no more than necessary. When asked to generate counterfactuals, we find that LLMs typically produce SCEs that are valid, but far from minimal, offering little insight into their decision-making behaviour. Worryingly, when asked to generate minimal counterfactuals, LLMs typically make excessively small edits that fail to change predictions. The observed validity-minimality trade-off is consistent across several LLMs, datasets, and evaluation settings. Our findings suggest that SCEs are, at best, an ineffective explainability tool and, at worst, can provide misleading insights into model behaviour. Proposals to deploy LLMs in high-stakes settings must consider the impact of unreliable self-explanations on downstream decision-making. Our code is available at github.com/HarryMayne/SCEs.

1 Introduction

Whether LLMs can reliably explain their decisions in natural language has recently become an area of intense research focus (Turpin et al., 2023; Parcalabescu and Frank, 2024). Self-explanation is increasingly seen as essential for effective human-computer interaction, allowing users to interrogate model decisions, compare reasoning steps against their prior beliefs, and determine whether behaviours are in line with their goals (Chen et al., 2025; Baker et al., 2025). One form of self-explanations are *self-generated counterfactual explanations* (SCEs), where a model explains its decision by modifying the input such that it would have predicted a different outcome (Madsen et al., 2024a; Dehghanighobadi et al., 2025).

Consider the deployment of LLMs to support clinical decision-making (Tu et al., 2025; Arora et al., 2025). A model might predict that a 60-year-old male with a systolic blood pressure of 135 mmHg is at high risk of developing heart disease. In response, a clinician might ask: *What would need to be different for the model to predict low risk instead?* The model could respond with a self-generated counterfactual explanation: *If the patient's blood pressure were* 110 *mmHg, I would have predicted low risk*. Such explanations highlight the features the model considers important, offer actionable insights for clinicians and patients, and reveal potential flaws in the model. Accordingly, counterfactuals are regarded as a particularly useful form of explanation (Miller, 2019; Wachter et al., 2017).

To serve as effective explainability tools, SCEs must satisfy the following criteria. First, SCEs should be *valid*: the revised input should change the model's prediction when re-evaluated in a new context window. Without validity, the explanation is a misleading representation of the model's counterfactual behaviour. Second, SCEs are often expected to be *minimal*: they should make the smallest edit required to change the outcome (Lewis, 1973; Wachter et al., 2017). By isolating the changes the model deems consequential, minimal counterfactuals provide clearer insight into the model's decision boundary. In many practical settings, minimality is preferable for satisfying the objectives of explainability (Lipton, 2018). For instance, in the heart disease example discussed, a minimal counterfactual would be more actionable for the patient (Keane and Smyth, 2020).

Prior work has shown that LLMs typically produce valid SCEs but has not addressed minimality (Dehghanighobadi et al., 2025; Randl et al., 2025). The question remains: Can LLMs gener-

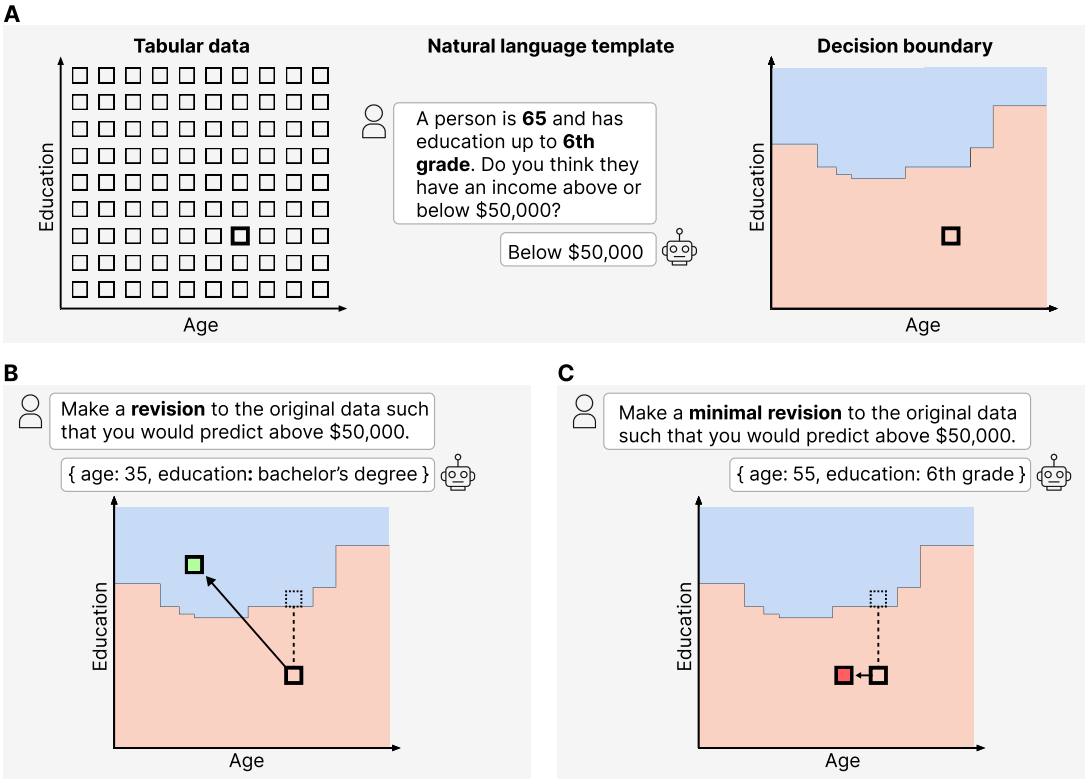

Figure 1: **Study design.** A. We evaluate models in tabular data, binary classification tasks. The model predictions form a decision boundary across the input space. B. We ask models to provide self-generated counterfactual explanations (SCEs) for their predictions. SCEs are *valid* when they cross the decision boundary (here, red → blue) and are *minimal* if they are close to the dashed instance at the decision boundary. When asked to provide counterfactual explanations, we find that SCEs are typically valid but far from minimal. C. In separate continuations from the original predictions, we ask models to provide minimal counterfactual explanations. In the majority of cases, these SCEs fail to cross the decision boundary. There is a trade-off between validity and minimality.

ate SCEs that are both valid and minimal? To answer this, we use tabular data prediction tasks. In these settings inputs have both natural language and tabular data representations, meaning that we can prompt LLMs in natural language, whilst measuring validity and minimality in structured tabular environments. Through experiments across several LLMs, datasets, and evaluation settings, we show the following:

Valid but not minimal. When prompted to generate counterfactual explanations, frontier LLMs achieve near-perfect validity. However, they typically make excessive changes to the original inputs, which trivially flip predictions. These SCEs provide little insight into the models' decision boundaries.

Minimal but rarely valid. When instructed to generate minimal counterfactuals, models generally make overly conservative edits that do not flip their predictions. Such SCEs misrepresent the models' true behaviour and could potentially lead to incorrect downstream decision-making if relied upon in high-stakes deployment. In this regard, LLMs do not know their own decision boundaries. As a consequence, even when using state-of-the-art LLMs, SCEs remain an unreliable, and potentially misleading, method of explaining model behaviour.

2 Self-generated counterfactual explanations

Consider a classification task where an LLM f takes natural language input x and predicts a binary output y. For a given input-output pair, a *counterfactual explanation* is an alternative input $x' \neq x$ that would lead to a specific alternative output $y' \neq y$ (Wachter et al., 2017). Typically, natural language counterfactuals are identified by algorithms that iteratively perturb keywords in the input (e.g., swapping "terrible" → "great" in a movie review), then test whether each perturbation changes the LLM's decision to the target output (Wang et al., 2024). In contrast, *self-generated counterfactual*

explanations occur when the LLM itself generates a candidate counterfactual to explain its own prediction (Madsen et al., 2024a; Dehghanighobadi et al., 2025). Whether an SCE flips the LLM's prediction can be tested by re-evaluating the LLM in a new context window. Independent evaluation is essential to avoid bias from the presence of the original prediction in context (Dehghanighobadi et al., 2025).

3 Methods

In this section, we describe our experiments to evaluate the properties of SCEs.

3.1 Study design

Our study design is outlined in Figure 1. The core innovation of our approach is to use tabular datasets $\mathcal{Z} = \{z_i\}_{i=1}^N$ containing numerical and ordinal features. We then use natural language templates ϕ to convert numerical tabular data z_i to natural language inputs x_i (see §A.1). As a result, we can prompt the LLMs in the natural language space, but evaluate the SCEs in the tabular data space. This creates a restricted input space, making it tractable to identify the closest point to the original input that would flip the model's prediction, and thus to measure the minimality of the SCEs.

All our datasets have discrete features, and contain all combinations of the feature values. As a result, the datasets are complete in the sense that they cover the full input space.

For each instance in a dataset, the LLM is first prompted to make a prediction $f(\phi(z_i)) = y$; then, given this prediction, it is prompted to revise the input such that it would have predicted the complementary class y'. Since the datasets are complete, eliciting predictions over the entire dataset is sufficient to determine the LLM's decision boundary over the input space (Figure 1A). The resulting decision boundary is then used to assess the properties of the SCE: validity (whether it crosses the decision boundary) and minimality (how much it crosses the boundary by) (Figure 1B-C).

3.2 Datasets

Our datasets are constructed as simplified versions of real-world datasets. In each case, we select two to four discrete features and enumerate all combinations of those features. The datasets are adapted from Cruz et al. (2024), Yasser (2022), and Janosi et al. (1989) (for more information see §A.1).

Income Given an individual's age and education level, the LLMs predict whether annual income exceeds $\$50,000$. LLMs are told that the data was collected in 2018 across the US. Age is numeric and education is ordinal ($N = 1,920$).

House prices Given the square-foot area of a house, the number of bedrooms, bathrooms, and floors, the LLMs predict whether the house price exceeds $\$1,500,000$. LLMs are told that the data was collected in 2015 across the US. All features are numeric ($N = 1,600$).

Heart disease Given an individual's age, sex, systolic blood pressure, and total cholesterol, the LLMs predict whether the individual has coronary heart disease. Age, systolic blood pressure, and total cholesterol are numeric, and sex is binary ($N = 1,936$).

3.3 Models

We evaluate four open-source and three proprietary LLMs: Gemma 2 27B (Gemma Team et al., 2024), Llama 3.3 70B (Grattafiori et al., 2024), DeepSeek-R1 Qwen 32B, DeepSeek-R1 Llama 70B (DeepSeek-AI et al., 2025), Claude Sonnet 3.7 (non-thinking) (Anthropic, 2025), GPT-4.1 (OpenAI, 2025a), and o3 (OpenAI, 2025b). Our main experiments evaluate all models at temperature 0, except o3, which has a fixed temperature of 1. We discuss the effect of non-zero temperature in §4.4.

3.4 Prompt settings

We evaluate SCEs generated under two prompt settings. First, an *unconstrained* setting, where models are simply asked to produce counterfactual explanations that would flip their predictions. This reflects how models behave by default when a user asks them to generate an SCE without any additional constraints. Second, a *minimal* setting, where models are instructed to make the smallest edits necessary to flip their predictions. The prompt wording specifies how minimality is defined and evaluated, ensuring that the models' notion of minimality aligns with our definition (see §3.5). To ensure that SCEs remain within the defined input space, we provide models with the feasible ranges of each feature. We also enforce a JSON schema with fixed output formats. The full prompts for each setting are provided in §A.3.

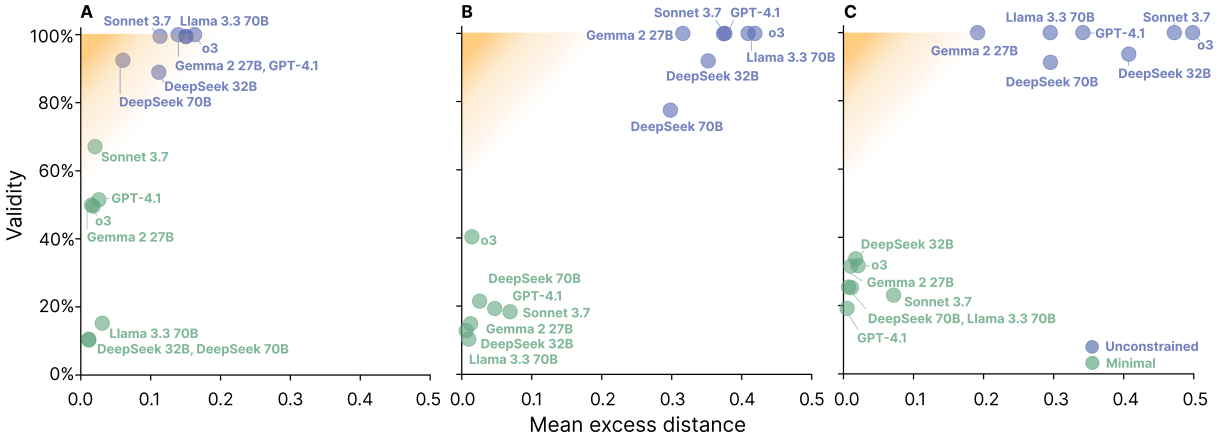

Figure 2: **SCE validity and minimality for income (A), house prices (B), and heart disease (C) datasets.** In the unconstrained prompting setting (purple), models typically provide valid SCEs, but they are far from minimal. In the minimal prompting setting (green), validity is notably lower, but, conditional on being valid, minimality is much better. No model can consistently satisfy both criteria across all three datasets. Orange regions indicate the direction of increasing validity and minimality (better performance).

3.5 Evaluation metrics

Validity For a tabular data instance z_i, where $f(\phi(z_i)) = y$, the tabular data representation of the SCE z_i' is *valid* if the model flips its decision when re-evaluated in a new context window, i.e. $f(\phi(z_i')) = y'$, where $y' \neq y$. Recall that an SCE is generated for every instance in a dataset. We define *Validity* (Val) as the proportion of dataset instances $z_i \in \mathcal{Z}$ for which the SCE successfully flips the model's prediction.

Minimality Any measure of minimality must be defined with respect to an underlying measure of distance. We use *Gower's Distance* (Gower, 1971), a simple pairwise distance function $d(z_i, z_j)$ which is defined over the tabular input space $\mathcal{Z}$ (see §A.4). This is preferable to more complex functions, e.g. Euclidean distance, as LLMs can reliably calculate it in-context (see §4.5). We define the *minimal counterfactual* as the closest point to the original input that would flip the LLM's decision

$$z_{i,\text{M}} \in \arg\min_{z_i' \in \mathcal{Z}} d(z_i, z_i'), \text{ s.t. } f(\phi(z_i')) = y'.$$

Given this, we define *Excess Distance (ED)* as the distance from the initial input to the SCE in excess of the distance to the minimal counterfactual,

$$\text{ED}_i = d(z_i, z_i') - d(z_i, z_{i,\text{M}}) \,.$$

We report the mean excess distance over all valid SCEs. This score ranges from 0 to 1, where lower scores are more minimal. Since the maximum Gower's Distance across the dataset instances is 1, this score can be interpreted as a fraction of the span of the dataset.

In addition, we define *Exact Match (EM)* as the proportion of dataset instances $z_i \in \mathcal{Z}$ for which the model exactly identifies a minimal counterfactual.

4 Results

4.1 Valid but not minimal

First we consider the unconstrained setting, where models are prompted to generate counterfactual explanations with no additional considerations. Here we find LLMs typically generate valid SCEs, but they are far from minimal (Figure 2). For example, o3 achieves 100% validity on all three datasets, but has mean excess distance scores of 0.16, 0.42, and 0.50 across the income, house prices and heart disease datasets, respectively. Given an excess distance of 1.0 is the distance from one extreme of the dataset to the other, a score of 0.16 can be interpreted as the SCEs overshooting the decision boundary by an average of 16% of the span of the dataset.

Similarly, Table 1 shows that models rarely identify the exact minimal counterfactual in this setting, with an average exact match across all models and datasets of only 2.89% (s.d. 5.64 percentage points).

All LLMs perform notably better in the income dataset. The exact reason for this is unclear, but potentially a result of this dataset being lower-dimensional (2 dimensions rather than 4).

Model	Unconstrained prompting			Minimal prompting		
	Income	House prices	Heart disease	Income	House prices	Heart disease
Gemma 2 27B	0.00	0.00	0.15	31.5	9.94	15.9
Llama 3.3 70B	3.33	0.00	0.00	8.80	7.31	10.9
DeepSeek-R1 32B	6.89	0.06	0.05	3.94	11.6	19.0
DeepSeek-R1 70B	19.7	0.19	0.42	4.22	13.2	17.3
Claude Sonnet 3.7	9.95	0.00	0.00	20.7	9.50	9.92
GPT-4.1	15.7	0.00	0.00	31.1	11.2	10.3
o3	4.27	0.00	0.00	26.2	23.9	17.5

Table 1: **Exact match: The percentage of SCEs that identify the minimal valid counterfactual across all instances in a dataset.** Models rarely identify the exact minimal counterfactual. Performance is higher in the minimal prompting setting, but no model exceeds 32%.

4.2 Minimal but rarely valid

Next we consider the minimal setting, where models are prompted to generate minimal counterfactual explanations. Here we observe a sharp drop in validity (Figure 2). In the income dataset, average validity falls from 97.14% (s.d. 4.49 pp) to 36.30% (s.d. 23.55 pp). However, conditional on being valid, the SCEs are more minimal than in the unconstrained prompting setting. For example, whilst o3 now only achieves validity around 40%, the mean excess distances of valid SCEs are 0.02, 0.01, and 0.02.

Similarly, Table 1 shows that models identify the minimal valid counterfactual more frequently than in the unconstrained setting, with an average exact match across all models and datasets of 14.9% (s.d. 8.01 pp).

These results suggest a trade-off between validity and minimality, where no model is able to reliably jointly satisfy both criteria. Surprisingly, there is no clear relationship between the general ability of models and their performance on this task, nor does there appear to be any benefit from leveraging inference-time compute to improve reasoning.

4.3 Where do models place SCEs?

In the unconstrained setting, we find LLMs often generate SCEs that trivially cross the decision boundary. This behaviour is most obvious in the house price dataset where there are clear monotonic relationships between each feature and average house price (more bedrooms always correspond to higher average price). In this case, models often generate SCEs which maximise or minimise every feature. For example, when modifying the input to predict a house priced above $\$1,500,000$, o3 returns the maximum point in the dataset (4 floors, 4 bathrooms, 5 bedrooms, area of $10,000$ sqft) in 46% of cases. We find similar behaviours in the heart disease dataset, e.g. o3 selects the maximum or minimum point in 34% of SCEs, but we do not see this pattern in the income dataset. This is potentially because age and education do not necessarily have monotonic relationships with income. However, we do find models favour specific regions. For example, when modifying the input to predict an income above $\$50,000$, Llama 3.3 70B often selects an age between 30 and 50 and education of a Bachelor's degree.

In the minimal setting, we find LLMs often make overly conservative edits which fail to cross the decision boundary. Figure 3 shows representative examples of this for Llama 3.3 70B in the income dataset. Occasionally, this strategy does produce SCEs which cross the decision boundary (if the initial input was close to the boundary), which explains why the SCEs are extremely minimal when valid, but rarely valid.

4.4 Robustness analysis

Here we show that the trade-off is robust to changes in the experimental design.

Distance function sensitivity We test three alternative distance functions: L_1 distance with each feature weighted by its inverse median absolute deviation, L_2 distance with each feature weighted by its inverse standard deviation (Wachter et al., 2017), and semantic distance, calculated as one minus the cosine similarity between inputs encoded using a sentence embedding model (Randl et al., 2025) (see §C.1 for details). We conduct these experiments with Llama 3.3 70B on the house price dataset since this is the dataset where the differ-

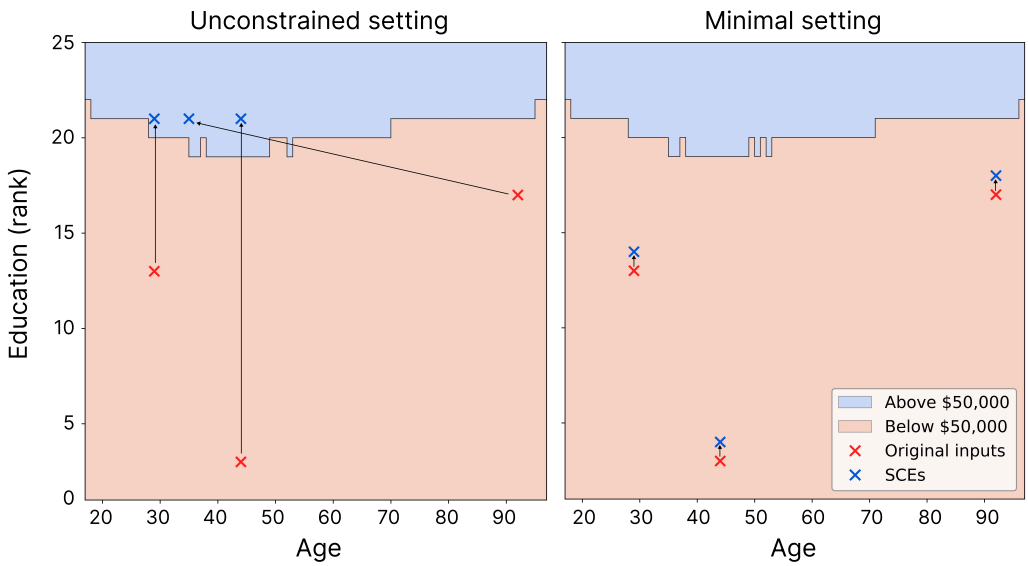

Figure 3: **Llama 3.3 70B's behaviour on the income dataset.** The three SCEs shown are randomly selected and are representative of the model's general behaviour. In the unconstrained prompting setting, all three SCEs cross the decision boundary and are therefore valid. However, the model makes an excessively large, non-minimal edit to the original input on the far right of the figure. In the minimal prompting setting, the model typically makes overly conservative edits which do not cross the decision boundary. Appendix Figure 10 shows the density of SCEs across all instances in the dataset.

ence between the two prompting settings is most pronounced.

Table 2 shows the results with excess distance scores normalised by the maximum distance across the dataset so that they fall in the $[0, 1]$ range. We find the validity-minimality trade-off occurs across all distance functions.

	Unconstrained		Minimal	
Distance	**Val**	**ED**	**Val**	**ED**
L_1	100	0.1403	12.66	0.0169
L_2	100	0.1026	11.87	0.0001
Semantic	100	0.5845	11.88	0.1280
Gower	100	0.1393	15.21	0.0305

Table 2: **Robustness to distance function choice.** The results are robust across three additional distance functions: L_1 distance with features weighted by the inverse median absolute deviation, L_2 distance with features weighted by the inverse standard deviation, and a semantic distance defined as one minus the cosine similarity between inputs when embedded by the `all-mpnet-base-v2` sentence embedding model. Mean excess distances are normalised by the maximum distance between two instances in the dataset. SCEs are generated using Llama 3.3 70B on the house price dataset. Val: Validity. ED: Mean excess distance.

Prompt sensitivity We test how sensitive our results are to the specific wording of the SCE-eliciting prompt. For both the unconstrained and minimal settings, we use o3 to generate twenty prompt variations, giving the model detailed instructions on how to modify the original wording (see §C.2). We then evaluate Llama 3.3 70B's performance on the house price dataset under each variation.

Figure 4 shows that performance remains consistent across prompt variations. Behaviour varies only slightly, and importantly, none of the prompts produce SCEs that are both valid and minimal.

Temperature We test whether our results hold when we use temperature 1.0 instead of greedy decoding. The results remain consistent: the validity–minimality trade-off persists (see full results in §C.3).

4.5 What limits performance?

Our results show that no model is able to satisfy both validity and minimality on any dataset. This raises the question: What limits performance? We consider three necessary but not sufficient criteria for success: (i) LLMs must have consistent decision boundaries, (ii) they must be able to operationalise the distance function, and (iii) they must be able to accurately self-predict their behaviour.

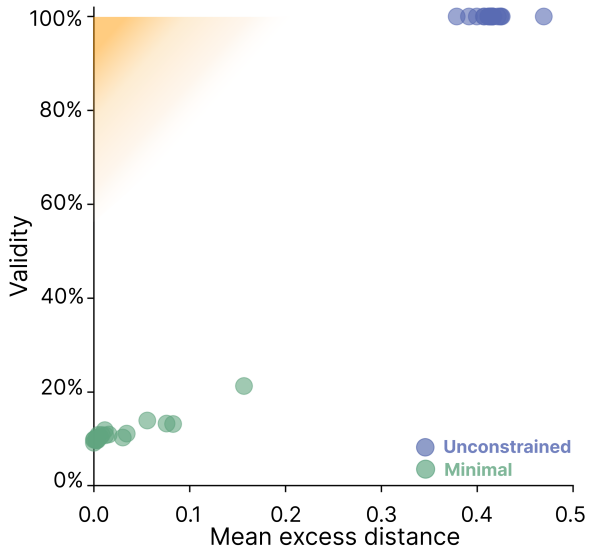

Figure 4: **Robustness to changes in the SCE-eliciting prompts.** We use o3 to generate 20 versions of both the unconstrained and minimal prompts, then evaluate Llama 3.3 70B's performance on the house price dataset under each variation. Behaviour is robust to the perturbations.

Decision boundary consistency Due to the stochasticity of LLMs, the decision boundaries for the prediction tasks may be inconsistent across different generations, making the task of identifying valid and minimal SCEs somewhat ill-defined. To test this, we use o3 to generate 50 versions of the income task prompt and evaluate the consistency of Llama 3.3 70B's predictions. We elicit predictions at temperature 1.0 to increase variation (see §D.1 for implementation details and example perturbations). We use the income dataset since we can visualise the decision boundary in two dimensions.

Figure 5 shows that the model's predictions are stable, except in a narrow region around the boundary. Importantly, we find that 91.4% of the original invalid SCEs remain invalid across all 50 versions of the decision boundary. This suggests that model failures cannot be attributed to decision boundary inconsistency.

Operationalising distance To select the closest valid SCE to the original input, models need to be able to compare candidate SCEs using the distance function. To do this correctly, they must be able to calculate the pairwise Gower's Distance between each candidate SCE and the original input. To test whether LLMs can do this, we conduct an experiment where models are given an initial instance from the house price dataset and four alternative

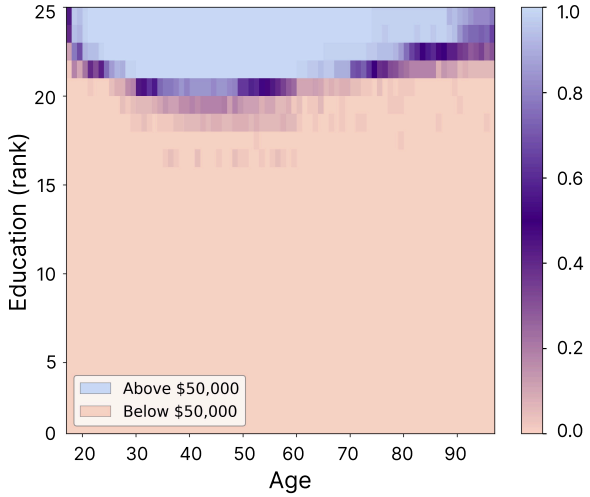

Figure 5: **Decision boundary consistency across 50 prompt perturbations.** Shading indicates the fraction of perturbations where Llama 3.3 70B predicted income above $\$50,000$ for an instance in the dataset. Predictions are elicited at temperature 1.0.

instances (all randomly selected). Their task is to identify the closest instance to the initial instance using Gower's Distance. We calculate accuracy over $1,000$ unique trials. Random performance is 25% (see §D.2 for details).

Table 3 shows a range in performance. Reasoning models trained to leverage inference-time compute perform notably better ($98 - 100\%$) than non-reasoning models ($58 - 72\%$). This reflects general trends in mathematical reasoning benchmarks (OpenAI, 2025b). The most capable model we consider, o3, scores 100%.

Model	**Accuracy** (%)
Gemma 2 27B	58.14
Llama 3.3 70B	71.92
DeepSeek-R1 32B	98.65
DeepSeek-R1 70B	99.79
Claude Sonnet 3.7	70.67
GPT-4.1	72.25
o3	100.00

Table 3: **Operationalising Gower's Distance.** The rate with which LLMs identify the closest point to an input point among four candidates using Gower's Distance. Accuracy is computed over $1,000$ trials with data coming from the house price dataset.

Since this experiment is more structured than the real task, we also analyse the distance calculations made by models' when producing the SCEs. By reviewing DeepSeek-R1 70B's reasoning traces,

we observe that the model typically approaches the task by first considering multiple candidate SCEs, then calculating the pairwise distance from the initial input to each of the candidates (see an example in Figure 6). In all cases reviewed, the model correctly calculates Gower's Distance in-context.

Along with the results of the multiple choice experiment, this suggests that the validity-minimality trade-off cannot solely be explained by an inability to operationalise the distance function. While this may be a limiting factor for the weaker, non-reasoning models, it cannot explain why state-of-the-art LLMs fail to produce high-quality SCEs.

Self-prediction Jointly satisfying validity and minimality requires LLMs to accurately predict how they would behave in alternative settings. This could be achieved either through perfect intrinsic knowledge of their decision boundaries or by iteratively testing candidate SCEs and reliably predicting whether each would flip their decision. In either case, we should expect to see evidence that models are attempting to predict their own behaviour, i.e. *self-prediction* (Premakumar et al., 2024).

To explore whether models spontaneously engage in self-prediction, we analyse DeepSeek-R1 70B's reasoning traces. While the extent to which chain-of-thoughts faithfully represent intrinsic reasoning remains a subject of debate (Chen et al., 2025; Barez et al., 2025), in practice they have proved to contain sufficient information about decision-making to serve as valuable explainability tools (Baker et al., 2025; Emmons et al., 2025).

By reviewing 30 randomly-selected reasoning traces from the house price dataset, we find that DeepSeek-R1 70B does not explicitly attempt to self-predict. Although the model shows a good understanding of minimality, it consistently fails to engage with the *self*-explanation aspect of the task. It frequently refers to the decision boundary of an external model, which it is unable to assess, and does not question whether the SCE would flip *its own* decision (see Figure 6). This is a fundamental necessary condition that is not met.

We also consider an experiment where we explicitly prompt models to self-predict in their reasoning traces. We instruct models to follow a six-step plan encouraging them to consider candidate SCEs, make explicit self-predictions and update their candidate to converge on the minimal valid counterfactual (see §E for details). Evaluating DeepSeek-R1 70B on the house price dataset we qualitatively observe the model making self-predictions in-context and having a better understanding of the self-explanation requirements of the task. Despite this, the aggregate results are largely unchanged with validity increasing from 21.44% to 23.34%, at the cost of excess distance increasing from 0.025 to 0.033. Anecdotally, we also observe that the self-predictions are often incorrect.

Whether accurate self-prediction is a fundamental limitation of LLMs is unclear and a question for future research. This would perhaps be unsurprising since standard pre-training techniques only incentivise acquisition of knowledge about the external world (Radford et al., 2018) and common post-training methods, including those using reinforcement learning to leverage inference-time compute (OpenAI, 2024; DeepSeek-AI et al., 2025), never reward models for accurately predicting how they would behave in independent context windows. In short, there is no optimisation pressure to develop a self-model (Premakumar et al., 2024). New learning objectives are likely required to incentivise accurate self-prediction.

5 Related work

We discuss two broad related areas: counterfactual explanations and self-explanations.

Counterfactual explanations Counterfactuals have been proposed as a way of explaining opaque models without requiring an understanding of their internal mechanisms (Wachter et al., 2017). In NLP settings, counterfactuals are typically generated using keyword-based perturbation methods (Wang et al., 2024) or using a separate language model to produce candidate counterfactuals (Wu et al., 2021; Nguyen et al., 2024).

Philosophical work has long suggested that counterfactual explanations should capture the *closest possible world* (Lewis, 1973), since this is often more actionable for the user. This sparked the importance of minimality in subsequent algorithmic research (Wachter et al., 2017), and led to the deployment of optimisation methods that minimise the distance between the original instance and the candidate counterfactual (Keane and Smyth, 2020). In NLP settings, the properties of counterfactuals have been well studied (Wang et al., 2024; Nguyen et al., 2024; McAleese and Keane, 2024). However, there is limited work measuring minimality in natural language domains. This is because, while it is easy to define *distance* in natural language spaces,

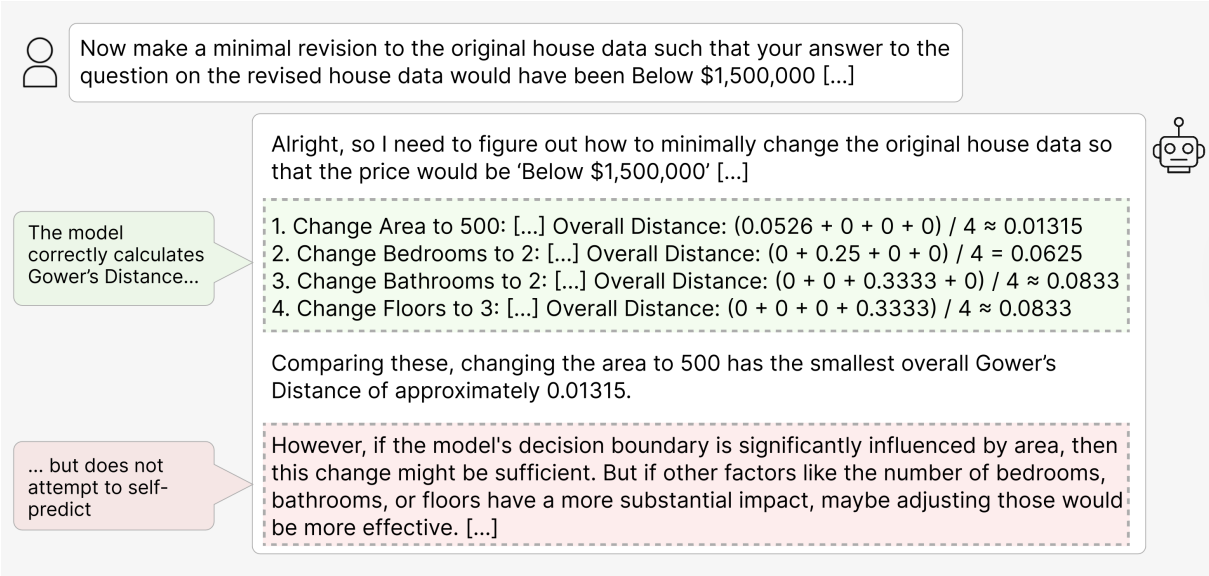

Figure 6: **DeepSeek-R1 70B can operationalise distance, but does not attempt self-prediction**. DeepSeek-R1 70B leverages Gower's Distance perfectly to compare changes, but does not appear to realise it is predicting its own behaviour. The reasoning traces are taken from an SCE generated on the house price dataset under the minimal prompting setting.

measuring *minimality* requires identifying the minimal counterfactual, which is computationally challenging given the input space is unconstrained. Our work addresses this by using tabular datasets to create restricted input spaces, allowing us to precisely measure minimality.

Self-explanations and SCEs As LLM capabilities have progressed, the ability of LLMs to offer explanations of their own decisions has emerged as a potential new paradigm for explainability (Madsen et al., 2024b). LLMs can provide plausible self-explanations either through their chain-of-thoughts or by explaining themselves post-hoc (Chen et al., 2024). Many studies implicitly treat these as interpretability tools (Barez et al., 2025); however, the critical question is whether these explanations are faithful representations of models' internal reasoning (Wiegreffe et al., 2021; Turpin et al., 2023; Lanham et al., 2023; Parcalabescu and Frank, 2024; Siegel et al., 2024; Barez et al., 2025).

An SCE is a model's attempt to self-explain using a counterfactual explanation. This is a structured form of self-explanation since there are additional constraints placed on the writing process compared to free-text explanations (Wiegreffe and Marasovic, 2021). Self-generated counterfactuals were first introduced by Madsen et al. (2024a), and later evaluated in subsequent works (Dehghanighobadi et al., 2025; Randl et al., 2025). These studies primarily focus on measuring validity across a range of open-ended generation settings. Our work builds on this by evaluating both validity and minimality, and by examining the trade-off between them.

6 Conclusion

Our findings reveal an important limitation of self-generated counterfactual explanations: a trade-off between validity and minimality. At best, SCEs are an ineffective means of explaining model behaviour, since LLMs naturally provide trivially valid counterfactuals. At worst, SCEs provide misleading insights into model behaviour, which could incorrectly steer human decision-making, potentially causing harm. Proposals to deploy LLMs in high-stakes settings must consider the impact of unreliable self-explanations.

Our study also investigates the factors that might limit performance, finding that models struggle to recognise the *self*-explanation elements of this task. Crucially, they do not spontaneously engage in self-prediction, a necessary condition for strong performance. Furthermore, performance does not significantly improve when models are explicitly prompted to self-predict in their reasoning traces, raising an important question about whether this is a fundamental limitation of LLMs.

7 Limitations

Alternative minimality metrics Our primary minimality metric is mean excess distance (§3.5). This metric is designed to capture the minimality of *valid* SCEs, but does not account for the properties of invalid SCEs. In some cases, it might be desirable to understand the distribution of these SCEs too. For example, we might wish to know whether the invalid SCEs lie just the wrong side of the decision boundary or are far removed. By introducing mean excess distance, we hope future work might design new metrics that incorporate both valid and invalid SCEs.

Self-prediction capabilities Our analysis suggests that models do not spontaneously engage in self-prediction when generating SCEs. We also show that aggregate performance is largely unchanged when we explicitly prompt LLMs to self-predict in their reasoning traces. However, we do not conduct detailed analysis into the self-prediction accuracy, make claims about whether LLMs are fundamentally capable of this, or the mechanisms by which they might hypothetically achieve this. These questions relate to the metacognitive capabilities of LLMs (Steyvers and Peters, 2025) and are promising directions for future work. When models are prompted or fine-tuned to self-predict within their chain-of-thoughts, can they do so accurately? If so, does SCE performance improve? Or are models fundamentally unable to accurately self-predict? Appendix E discusses these thoughts further.

Generalisation to real-world datasets Our experiments use three synthetic datasets, designed as simplified versions of real-world datasets. This is central to our methodology, as it allows us to generate datasets which contain every combination of the discrete input features. As a result, the exact minimum counterfactual can be located, enabling the measurement of SCE minimality. An additional benefit of using toy datasets is that they provide the simplest possible environment in which models might identify valid and minimal counterfactuals. Across all experiments, we use datasets with two to four input dimensions. Since models fail in these datasets, we anticipate that they will also struggle in more complex, higher-dimensional datasets, where locating a valid and minimal counterfactual is significantly harder.

Acknowledgements

We thank the reviewers for their helpful feedback. We also thank Jabez Magomere and Jakob Foerster for insightful conversations and feedback during the research process. Harry Mayne's PhD is supported by the Economic and Social Research Council grant ES/P000649/1. Andrew Bean's PhD is supported by the Clarendon Fund Scholarships at the University of Oxford. Yushi Yang's PhD is supported by the Oxford Internet Institute and the Dieter Schwarz Foundation Project Award. Ryan Othniel Kearns was supported by a Fellowship from the Cosmos Institute.

References

Anthropic. 2025. Claude 3.7 sonnet and claude code. https://www.anthropic.com/news/claude-3-7-sonnet.

Rahul K. Arora, Jason Wei, Rebecca Soskin Hicks, Preston Bowman, Joaquin Quiñonero-Candela, Foivos Tsimpourlas, Michael Sharman, Meghan Shah, Andrea Vallone, Alex Beutel, Johannes Heidecke, and Karan Singhal. 2025. Healthbench: Evaluating large language models towards improved human health. *Preprint*, arXiv:2505.08775.

Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. 2025. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. *Preprint*, arXiv:2503.11926.

Fazl Barez, Tung-Yu Wu, Iván Arcuschin, Michael Lan, Vincent Wang, Noah Siegel, Nicolas Collignon, Clement Neo, Isabelle Lee, Alasdair Paren, Adel Bibi, Robert Trager, Damiano Fornasiere, John Yan, Yanai Elazar, and Yoshua Bengio. 2025. Chain-of-thought is not explainability. *Preprint*, alphaXiv:2025.02v1. AlphaXiv preprint, version 1.

Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, and Ethan Perez. 2025. Reasoning models don't always say what they think. *Preprint*, arXiv:2505.05410.

Yanda Chen, Ruiqi Zhong, Narutatsu Ri, Chen Zhao, He He, Jacob Steinhardt, Zhou Yu, and Kathleen Mckeown. 2024. Do models explain themselves? Counterfactual simulatability of natural language explanations. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 7880–7904. PMLR.